

COUNTEREXAMPLES TO GYODA'S INJECTIVITY CONJECTURE FOR GENERALIZED MARKOV TREES

ABSTRACT. Gyoda conjectured that, in each fixed generalized Markov tree, equality of numerical labels $n_s = n_t$ forces equality of the corresponding Farey labels. We refute this conjecture by the collision

$$n_{1/4} = n_{2/3} = 889 \quad \text{in} \quad \text{MT}(22, 0, 5, \text{id}),$$

whose parameter triple is pairwise distinct. We also construct infinitely many counterexample trees, beginning with $n_{1/5} = n_{2/3} = 89$ in $\text{MT}(6, 0, 0, \text{id})$. In all of these examples the equal numerical labels have different coordinate-position tags; thus the result concerns the map $t \mapsto n_t$, exactly as in the stated conjecture.

1. THE CONJECTURE AND THE RESULT

For $k_1, k_2, k_3 \in \mathbb{Z}_{\geq 0}$, the generalized Markov equation is

$$(1) \quad x^2 + y^2 + z^2 + k_1 yz + k_2 zx + k_3 xy = (3 + k_1 + k_2 + k_3)xyz.$$

Gyoda and Matsushita introduced its mutation structure in [3]. Gyoda subsequently synchronized the resulting binary trees with the Farey tree and formulated the following assertion as Conjecture 7.6 in [2], where he recorded that no counterexample was known: for every fixed tree $\text{MT}(k_1, k_2, k_3, \sigma)$ and irreducible fractions $s, t \in [0, 1]$,

$$n_s = n_t \quad \implies \quad s = t.$$

Here (n_t, i_t) denotes the generalized Markov number and its coordinate position at the node whose middle Farey label is t . A different generalization of the uniqueness conjecture for the same equation (1), stated for size-ordered triples rather than for the labeling $t \mapsto n_t$, had been proposed shortly before by Chen and Jia [1, Conjecture 8.2]; it is delimited from Conjecture 7.6 in the Status and scope section below.

For completeness, we recall the part of the construction used below. The Farey tree has root $(0/1, 1/1, 1/0)$, and the children of a Farey triple (r, t, s) are

$$(r, r \oplus t, t) \quad \text{and} \quad (t, t \oplus s, s), \quad \frac{a}{b} \oplus \frac{c}{d} := \frac{a+c}{b+d}.$$

For $\sigma = \text{id}$, the corresponding generalized Markov tree has root

$$((1, 1), (k_2 + 2, 2), (1, 3)).$$

2020 *Mathematics Subject Classification.* 11D25, 11B39.

Key words and phrases. generalized Markov equation, generalized Markov tree, Farey labeling, uniqueness conjecture, Fibonacci numbers.

If $v = ((a, h), (b, i), (c, j))$, its left and right children are

$$L(v) = \left((a, h), \left(\frac{a^2 + k_j ab + b^2}{c}, j \right), (b, i) \right),$$

$$R(v) = \left((b, i), \left(\frac{b^2 + k_h bc + c^2}{a}, h \right), (c, j) \right).$$

The middle pair at the node corresponding to (r, t, s) is (n_t, i_t) .

Theorem 1. *Gyoda's Conjecture 7.6 is false.*

(1) In $\text{MT}(22, 0, 5, \text{id})$,

$$(n_{1/4}, i_{1/4}) = (889, 3), \quad (n_{2/3}, i_{2/3}) = (889, 1).$$

In particular, the conjecture fails for a pairwise distinct parameter triple.

(2) Let $F_0 = 0$, $F_1 = 1$, and $F_{r+1} = F_r + F_{r-1}$ for $r \geq 1$. For every $m \geq 0$, set

$$d_m = 30m + 5, \quad q_m = \frac{F_{60m+11} - 29}{10}.$$

Then $q_m \in \mathbb{Z}_{\geq 0}$ and, in $\text{MT}(q_m, 0, 0, \text{id})$,

$$(n_{1/d_m}, i_{1/d_m}) = (F_{60m+11}, 2), \quad (n_{2/3}, i_{2/3}) = (F_{60m+11}, 1).$$

For $m = 0$, this is the collision $n_{1/5} = n_{2/3} = 89$ in $\text{MT}(6, 0, 0, \text{id})$.

2. A PAIRWISE DISTINCT COUNTEREXAMPLE

Take $(k_1, k_2, k_3) = (22, 0, 5)$. Starting at the root, three left mutations give

$$\begin{aligned} ((1, 1), (2, 2), (1, 3)) &\xrightarrow{L} ((1, 1), (15, 3), (2, 2)) \\ &\xrightarrow{L} ((1, 1), (113, 2), (15, 3)) \\ &\xrightarrow{L} ((1, 1), (889, 3), (113, 2)), \end{aligned}$$

where

$$15 = 1^2 + 5(1)(2) + 2^2, \quad 113 = \frac{1^2 + 15^2}{2},$$

$$889 = \frac{1^2 + 5(1)(113) + 113^2}{15}.$$

The middle Farey labels along this path are $1/2$, $1/3$, and $1/4$. On the other hand, the right child of the $1/2$ node is

$$R(((1, 1), (15, 3), (2, 2))) = ((15, 3), (889, 1), (2, 2)),$$

$$889 = 15^2 + 22(15)(2) + 2^2,$$

whose middle Farey label is $2/3$. Hence

$$(n_{1/4}, i_{1/4}) = (889, 3), \quad (n_{2/3}, i_{2/3}) = (889, 1).$$

The position tags correspond to the ordered solutions $(x, y, z) = (1, 113, 889)$ and $(x, y, z) = (889, 2, 15)$ of

$$x^2 + y^2 + z^2 + 22yz + 5xy = 30xyz,$$

as substitution verifies:

$$\begin{aligned} 1 + 113^2 + 889^2 + 22(113)(889) + 5(1)(113) &= 3013710 = 30(1)(113)(889), \\ 889^2 + 2^2 + 15^2 + 22(2)(15) + 5(889)(2) &= 800100 = 30(889)(2)(15). \end{aligned}$$

This proves Theorem 1(1).

One remark on the pair, used nowhere in the proofs. The two colliding labels lie at different depths of the Farey tree, $1/4$ at depth 3 and $2/3$ at depth 2, but they have the same weight, that is, the same sum of numerator and denominator, $1+4 = 2+3 = 5$; and $\{1/4, 2/3\}$ is the first pair of distinct irreducible fractions in $[0, 1]$ of equal weight. A search for collisions that enumerates Farey labels by weight rather than by depth therefore reaches this pair first.

3. AN INFINITE FAMILY

Lemma 2. *Fix $q \in \mathbb{Z}_{\geq 0}$. In $\text{MT}(q, 0, 0, \text{id})$,*

$$n_{1/d} = F_{2d+1} \quad (d \geq 1), \quad n_{2/3} = 29 + 10q.$$

Moreover, $i_{1/d} = 2$ for odd d and $i_{1/d} = 3$ for even d , while $i_{2/3} = 1$.

Proof. Write $u_d = n_{1/d}$ and $i_d = i_{1/d}$. Since $k_2 = 0$, the root is

$$((1, 1), (2, 2), (1, 3)),$$

whose middle Farey label is $1/1$, so $u_1 = 2$ and $i_1 = 2$. Its left child is

$$\begin{aligned} L(((1, 1), (2, 2), (1, 3)))) &= \left((1, 1), \left(\frac{1^2 + k_3(1)(2) + 2^2}{1}, 3 \right), (2, 2) \right) \\ &= ((1, 1), (5, 3), (2, 2)), \end{aligned}$$

with middle Farey label $1/2$; hence $u_2 = 5$ and $i_2 = 3$.

Along the left spine the first entry stays $(a, h) = (1, 1)$, so the node labeled $1/d$ is $((1, 1), (u_d, i_d), (u_{d-1}, i_{d-1}))$ and its left child is

$$\left((1, 1), \left(\frac{1 + k_{i_{d-1}}u_d + u_d^2}{u_{d-1}}, i_{d-1} \right), (u_d, i_d) \right).$$

The new entry therefore inherits the tag of the third component of its parent, and all tags occurring along the spine lie in $\{2, 3\}$, where $k_2 = k_3 = 0$. Hence

$$u_{d+1} = \frac{u_d^2 + 1}{u_{d-1}}, \quad i_{d+1} = i_{d-1} \quad (d \geq 2),$$

so the position tags alternate: with $i_1 = 2$ and $i_2 = 3$ we get $i_{1/d} = 2$ for odd d and $i_{1/d} = 3$ for even d . The Fibonacci identity

$$F_{2d-1}F_{2d+3} = F_{2d+1}^2 + 1,$$

which is the case $n = 2d + 1$, $r = 2$ of Catalan's identity $F_n^2 - F_{n-r}F_{n+r} = (-1)^{n-r}F_r^2$, together with $u_1 = 2 = F_3$ and $u_2 = 5 = F_5$, therefore proves by induction that $u_d = F_{2d+1}$.

The node labeled $1/2$ is $((1, 1), (5, 3), (2, 2))$. Its right child is

$$\left((5, 3), \left(\frac{5^2 + k_1(5)(2) + 2^2}{1}, 1 \right), (2, 2) \right) = ((5, 3), (29 + 10q, 1), (2, 2)),$$

which proves the remaining assertions. $\square$

Proof of Theorem 1(2). A direct iteration of the Fibonacci recurrence modulo 10 gives

$$(F_{60}, F_{61}) \equiv (0, 1) \pmod{10}.$$

The recurrence then implies $F_{r+60} \equiv F_r \pmod{10}$ for every $r \geq 0$. Hence

$$F_{60m+11} \equiv F_{11} = 89 \equiv 9 \pmod{10},$$

so q_m is an integer. Since $F_{60m+11} \geq F_{11} = 89$, one also has $q_m \geq 6$.

Because $2d_m + 1 = 60m + 11$, Lemma 2 yields

$$n_{1/d_m} = F_{60m+11} = 29 + 10q_m = n_{2/3}.$$

Furthermore, d_m is odd, so the corresponding position tags are 2 and 1. Finally, $d_m \geq 5$, and hence $1/d_m \neq 2/3$. Since q_m is strictly increasing, these are infinitely many distinct counterexample trees. $\square$

STATUS AND SCOPE

The equal numerical labels in Theorem 1 have different position tags. Thus the theorem refutes Conjecture 7.6 exactly as stated, namely the injectivity of $t \mapsto n_t$; it does not address injectivity of $t \mapsto (n_t, i_t)$. Part (2) gives infinitely many counterexample trees, not infinitely many collisions in one fixed tree. No claim is made about global minimality, the classical case $(0, 0, 0)$, or the symmetric locus $k_1 = k_2 = k_3$.

Theorem 1 is also not a restatement of an already known failure. Question 7.3 of [2] — whether a generalized Markov number determines the generalized Markov triple in which it is the largest member — is the direct transcription of Frobenius's uniqueness conjecture, and [2] records that it is already answered in the negative; Conjecture 7.6 is the generalization of an equivalent reformulation that Gyoda states instead, and for which he records that no counterexample was known. What Theorem 1(1) exhibits is a collision inside a single tree. The nodes at $1/4$ and $2/3$ of $\text{MT}(22, 0, 5, \text{id})$ are distinct, namely $((1, 1), (889, 3), (113, 2))$ and $((15, 3), (889, 1), (2, 2))$, yet 889 stands in the second of the three components at each — the component recorded by the Farey label — so that $n_{1/4} = n_{2/3} = 889$. Their position tags present the two nodes as the ordered solutions $(1, 113, 889)$ and $(889, 2, 15)$ of (1) at $(k_1, k_2, k_3) = (22, 0, 5)$. In each of these 889 is the largest entry in size, while its coordinate position is 3 in the first and 1 in the second; and the

ordering is not free, since of the six orderings of 2, 15, 889 only (889, 2, 15) satisfies the equation. That is precisely what Conjecture 7.6 forbids.

A different generalization of the uniqueness conjecture for equation (1) was proposed, under the name *Generalized Uniqueness Conjecture*, by Chen and Jia [1, Conjecture 8.2], posted on 5 November 2025 and so before [2]; Section 8.2 of [1] develops a heuristic method for locating counterexamples to it. That conjecture is a statement about triples ordered by size rather than about the labeling $t \mapsto n_t$ of a fixed tree, and (1) is not symmetric in x, y, z unless $k_1 = k_2 = k_3$: the two solutions exhibited in Section 2 occupy different coordinate positions, (1, 113, 889) having its largest entry in position 3 and (889, 2, 15) in position 1. We make no claim about [1, Conjecture 8.2] and do not address it here.

We have not found a previous refutation of Conjecture 7.6, nor a previous appearance of the collision at 889: a search of the generalized Markov literature, including [1, 2, 3], turned up the conjecture itself and the heuristic search method of [1, Section 8.2], but no counterexample. That search was not exhaustive and has not been through external review, so any earlier refutation we have missed takes precedence over this note.

Exact verification. Every assertion above is a finite computation with integers, and the data printed here fix it completely: the four mutation evaluations of Section 2 — 15, 113 and 889 along the left spine, and 889 at the right child of the $1/2$ node — the two substitutions displayed there, and, for Theorem 1(2), Lemma 2 together with the congruence $F_{r+60} \equiv F_r \pmod{10}$. All arithmetic is exact integer arithmetic, each division performed by a mutation being exact; no floating-point arithmetic and no randomness are involved. It was recomputed once independently from that data alone — the six checks of Section 2, and, for each m with $0 \leq m \leq 10$, the integrality of q_m and the two labels n_{1/d_m} and $n_{2/3}$ obtained by mutating $\text{MT}(q_m, 0, 0, \text{id})$ from its root instead of quoting Lemma 2 — and every check agreed with what is reported here. The program and its transcript are retained in an auxiliary archive and are available on request; no part of the verification depends on them beyond what is printed above.

REFERENCES

- [1] Z. Chen and Z. Jia, *Tropicalization and cluster asymptotic phenomenon of generalized Markov equations*, arXiv:2511.03428v2 [math.NT] (2025).
- [2] Y. Gyoda, *Generalized discrete Markov spectra*, arXiv:2512.04547v4 [math.NT] (2025), revised 2026.
- [3] Y. Gyoda and K. Matsushita, *Generalization of Markov Diophantine equation via generalized cluster algebra*, Electron. J. Combin. **30** (2023), no. 4, Paper No. P4.10, doi:10.37236/11420.